DAY 9 — Data Visualization with Matplotlib & Seaborn

Day 9 was a major milestone in developing data storytelling skills. Visualization converts raw numbers into meaningful insights that are easy to interpret. Using Matplotlib and Seaborn, we created different types of informative charts.

🔹 1. Introduction to Matplotlib

Matplotlib is the oldest and most powerful library for plotting in Python.

We practiced:

plt.plot()

plt.bar()

plt.hist()

plt.pie()

plt.show()

Understanding parameters like:

- figsize
- linewidth
- markers
- titles
- labels
  helped improve the clarity of plots.

🔸 2. Introduction to Seaborn

Seaborn is built on top of Matplotlib and is used for statistical charts.

We plotted:

- countplot
- jointplot
- pairplot
- heatmap
- violin plot
- swarm plot

Seaborn automatically adds better colors, themes, and statistics.

🔸 3. Visualizing Relationships

We created:

- Scatter plots
- Regression lines
- Distribution plots

These helped understand correlation between features.

🔸 4. Visualizing Categorical Data

We used:

- bar plot
- count plot
- box plot

These visualizations helped analyze gender distribution, category counts, and salary comparisons.

🔸 5. Time-Series Visualization

Since our major project was Tesla stock prediction, this topic was extremely important.

We visualized:

- Daily closing prices
- Trends over months
- Seasonal patterns

Using:

```
plt.plot(df['Date'], df['Close'])
```

🔸 6. Heatmaps & Correlation Matrix

A heatmap quickly shows:

- How much two features are related
- Which features are important for ML
- Which columns do not matter

This helps in feature selection.

Key Takeaways from Day 9

- Visualization is essential for storytelling
- Matplotlib is flexible, Seaborn is easier for statistics
- Heatmaps reveal hidden patterns
- Time-series plots help detect trends and seasonality

Visualization made me connect deeply with the dataset and improved my analytical skills.